Winning Patterns: Comparative Analysis of Hackathon Submissions and GitHub Repositories

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***Abstract*—abcd**

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I. Introduction

Hackathon

II. Relevance and Motivation

III. Literature Review

IV. Data Collection and Preprocessing

The hackathon project data was collected from Devpost.com [X]. For this study, we selected a total of 50 hackathons, distributed evenly across five years (2020–2024), with 10 hackathons per year. The hackathons were randomly chosen from a pool of Major League Hacking (MLH) events [X] and other popular hackathons. A detailed workflow to gather the data is shown below in Figure 1.

Figure 1: Hackathon Data Collection and Organization Process

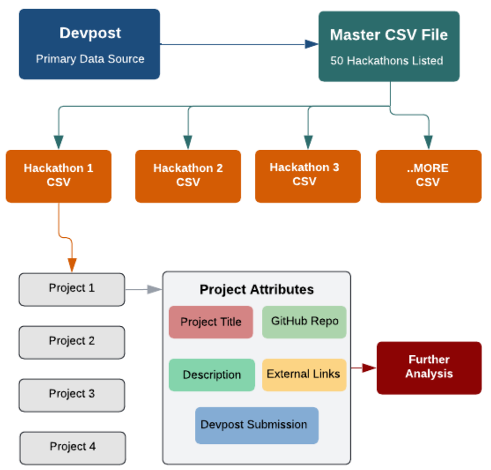
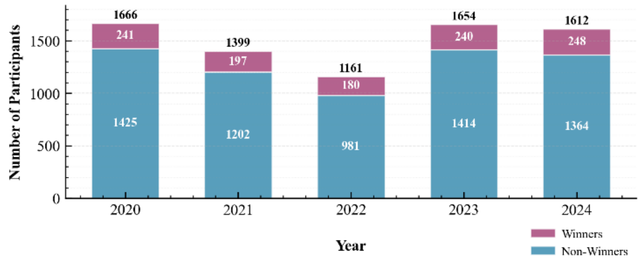


Figure 1: Workflow of Hackathon Data Collection and Organization. Data from Devpost is compiled into a master CSV, segmented into individual hackathon CSVs, detailed by project attributes, and finally analyzed further to collect more insights.

For data collection, we utilized BeautifulSoup [X], a Python library designed for web scraping and content extraction. The workflow, summarized in Figure 1, began with the construction of a master CSV file containing the list of hackathons included in this study along with links to their respective project pages. From this index, we generated hackathon-specific CSV files, each capturing all submitted projects for the corresponding event. For every project entry, we extracted comprehensive metadata including the project title, description, team composition, and external resources such as GitHub repositories, video demonstrations, and deployed applications. These features were systematically organized as columns within each hackathon dataset, thereby producing a structured and multimodal corpus suitable for subsequent analysis.

Figure 2: Hackathon Participation and Success Rates Over Time



V. Tech Stack and Repository Analysis

VI. Topic Modelling

VII. Discussion

VIII. Further Works

IX. References

[1].